AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 18 · STUDENT LESSON

Error Analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 15: errors can be grouped by relevant conditions.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Categorize development errors using evidence. Try the recall before opening notes.

Your learning target

I can categorize development errors using **evidence**.

I can design a controlled change with a stated **hypothesis**.

I can preserve failures and test boundaries.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

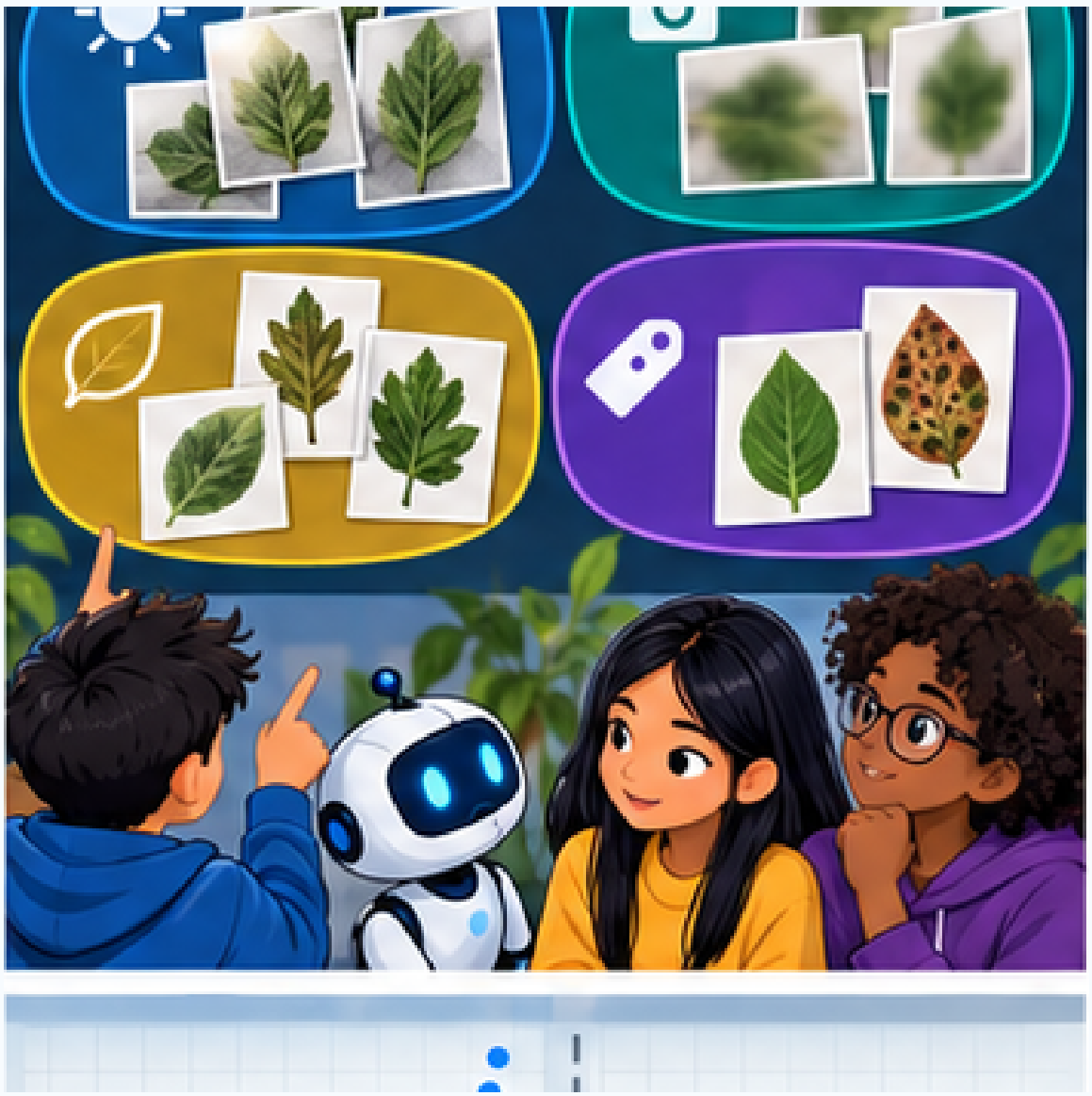
Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.



Your target

LEARNING TARGET A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

Builder name and experiment date:

Readiness for the connected explanation

What does a confusion matrix omit?

This week's end goal

By the end, I can explain **error analysis** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

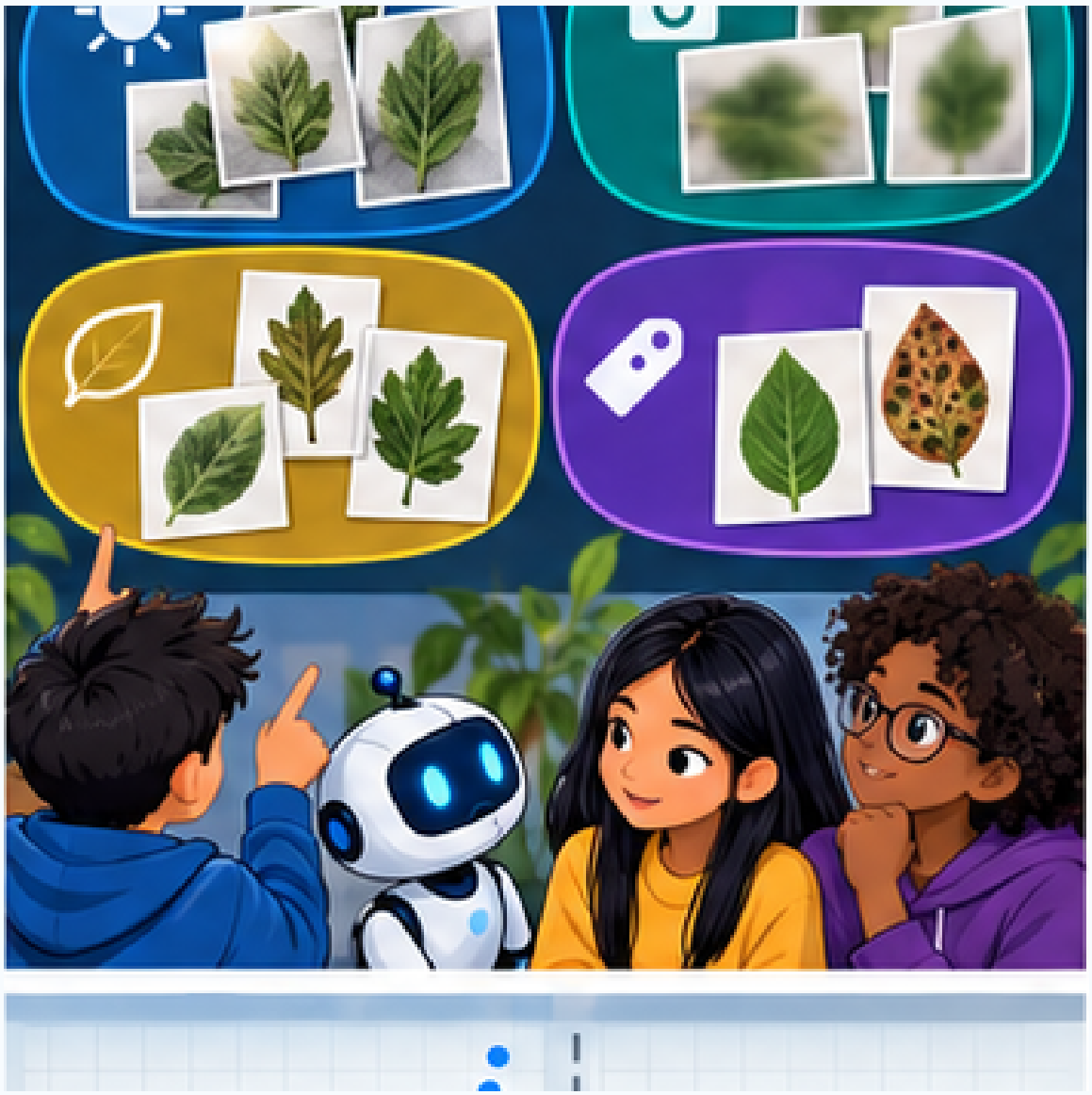
Notice the evidence

> Investigate the supplied Error Analysis case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **error analysis** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Error analysis begins with **prediction** IDs and reference labels. Inspect features, collection conditions and the labelling policy. Do not assume every surprising **prediction** proves the reference is wrong.

Use development errors to propose a change and specify what outcome would support the explanation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **error analysis** and controlled follow-up.

STUDENT LESSON · 3 OF 18

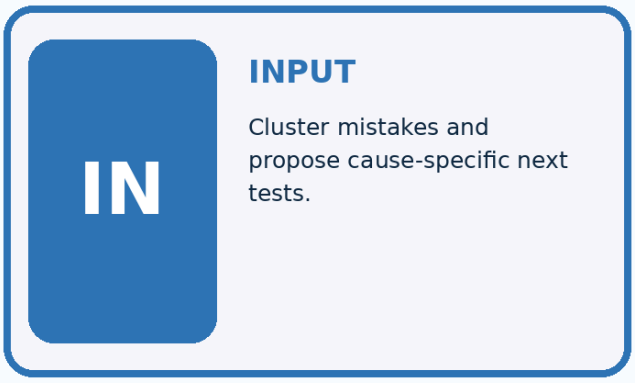
See how it works

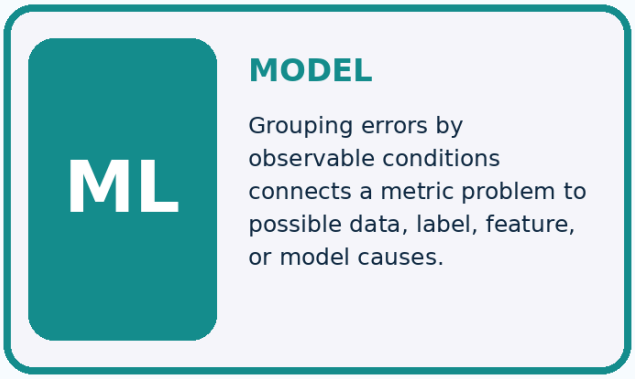
> Follow the mechanism for Error Analysis. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Cluster mistakes and propose cause-specific next tests.

MODEL IDEA Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.

PROOF NEEDED Use an unseen case, visible calculation, **error analysis**, and a stated limitation.

How the mechanism works

UNDER THE HOOD Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species. Largest cluster suggests a lighting representation problem, not proof of the cause.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

A validation audit has 12 dim-light cases with 4 errors and 18 bright-light cases with 3 errors. Rates are 4/12=1/3 and 3/18=1/6. Dim lighting is a useful lead, not proof of causation: camera, route or labels may also differ.

A planned change uses a **training**-fitted brightness normalization while keeping split and classifier settings fixed.

```
old_errors = [4, 3]
new_errors = [2, 4]
counts = [12, 18]
for old, new, n in zip(old_errors, new_errors, counts):
    print(old / n, new / n)
print(sum(old_errors), sum(new_errors))
# Total errors fall from 7 to 6; one group worsens.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

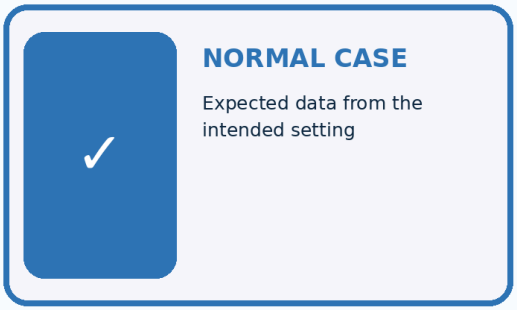
Compare the cases

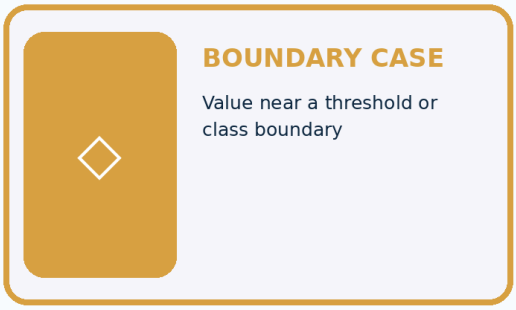
> Compare the supplied cases for Error Analysis. Identify the detail that changes the conclusion.

EVALUATION DESIGN

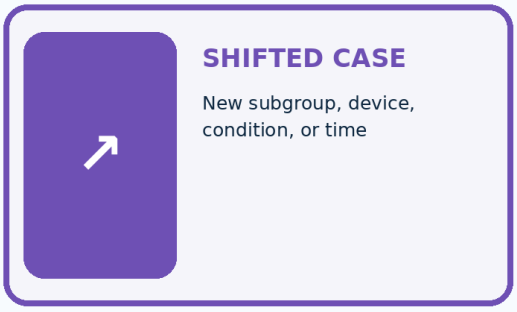
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Dim errors fall to1/6 in a later same-case replay.

Predicted result

Prediction: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

Why the result follows

Reason: Observed improvement on reused cases is not an untouched final generalisation test.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Error Analysis to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
ERROR SLICE
subset sharing a condition
2
FAILURE CLUSTER
related mistakes with a likely cause
3
ROOT-CAUSE HYPOTHE
testable explanation for errors
4
CONTROLLED REPAIR
one change aimed at one cause

Term	Precise meaning	My ML example
error slice	subset sharing a condition	
failure cluster	related mistakes with a likely cause	
root-cause hypothesis	testable explanation for errors	
controlled repair	one change aimed at one cause	

MISCONCEPTION TO REPAIR Looking closely at one surprising error is enough to diagnose the model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Error analysis	Inspection of mistakes to propose explanations
Hypothesis	A testable proposed explanation
Controlled comparison	A comparison that changes one planned factor while holding relevant conditions fixed
Regression	A new failure introduced by a change

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Error Analysis, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 18 • WORKED EVIDENCE

1. Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species.
2. Largest cluster suggests a lighting representation problem, not proof of the cause.
3. Collect or transform low-light examples, changing one factor.
4. Rerun the same evaluation and inspect other slices for regressions.

Step 1

STEP 1 Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species.

Step 2

STEP 2 Largest cluster suggests a lighting representation problem, not proof of the cause.

Step 3

STEP 3 Collect or transform low-light examples, changing one factor.

Step 4

STEP 4 Rerun the same evaluation and inspect other slices for **regression**s.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Error analysis and controlled follow-up at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Inspect	IDs and conditions	Propose a hypothesis
Change	One planned factor	Keep comparison fixed
Evaluate	All validation records	Report gains and regressions

After the change, the same validation groups have 2/12 and 4/18 errors. Total errors fall from 7 to 6, but bright-light errors increase from 3 to 4. Report both results. A net improvement does not mean every part improved, and repeated experiments can overfit validation.

Does the difference prove dim light causes model errors? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Error Analysis

Find an error cluster before choosing a controlled follow-up.

The facts we will use

Supplied facts

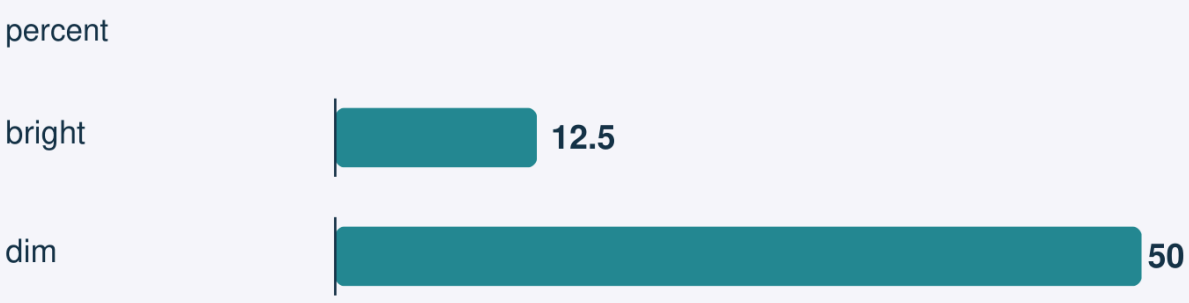
Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/18

Errors per 100 photos in each lighting slice



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/18

Error Analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice errors

Bright: 12.5%; dim: 50%. Total errors4/14, about28.6%.



2. Inspect dim failures

Check blur, lighting and label uncertainty instead of assuming lighting caused every error.



3. Plan controlled pair

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

CHANGE ONE THING

Dim errors fall to1/6 in a later same-case replay.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute slice errors	Bright: 12.5%; dim: 50%. Total errors4/14, about28.6%.
2. Inspect dim failures	Check blur, lighting and label uncertainty instead of assuming lighting caused every error.
3. Plan controlled pair	Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

Keep the stated limit

Read the labelled picture: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Dim errors fall to1/6 in a later same-case replay. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Error Analysis claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Looking closely at one surprising error is enough to diagnose the model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Looking closely at one surprising error is enough to diagnose the model.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Do not delete difficult cases to raise the score or reuse final-test errors for an unreported revision.

Once test feedback guides a change, treat it as development **evidence** and obtain a new independent evaluation for a fresh final claim.

> **A team removes all disputed records only from the new model report. Why is the comparison invalid?**

STUDENT LESSON · 8 OF 18

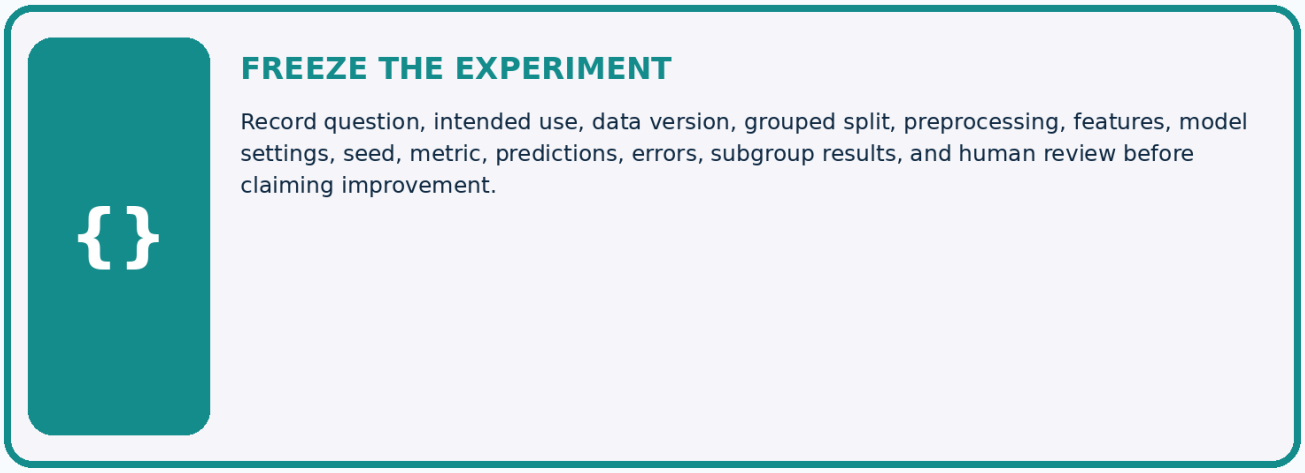
Practise together

> Practise Error Analysis with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Write a change record: **hypothesis**, exact code/data version, controlled factors, predicted effect, full before/after results and decision. Review disagreements using a consistent annotation policy.

If labels are corrected, version the data and rerun all candidates on the same corrected evaluation set.

Practise with less help: Explain the limit

Use workbook W2 for the connected example "**Error analysis** and controlled follow-up". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Error analysis** and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Error Analysis. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Error Clinic: annotate every wrong prediction, cluster by condition, write hypotheses, and select one measurable next test.

1. Write the **hypothesis** and expected **evidence** before running.

2. Error Clinic: annotate every wrong **prediction**, cluster by condition, write hypotheses, and select one measurable next test.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

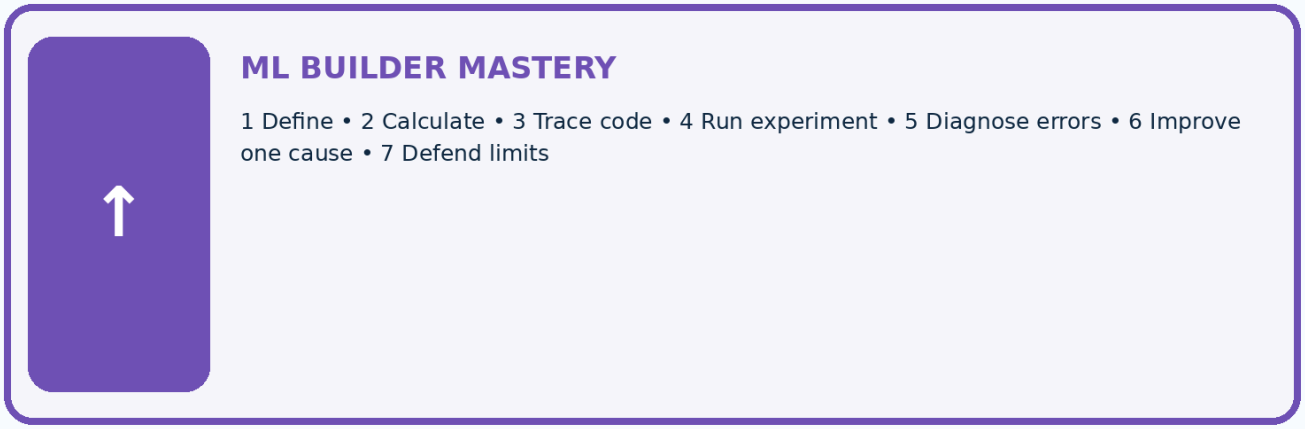
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Error Analysis. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **error slice**, **failure cluster**, **root-cause hypothesis**, **controlled repair**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Looking closely at one surprising error is enough to diagnose the model.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

THIS WEEK'S CHECKPOINT Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

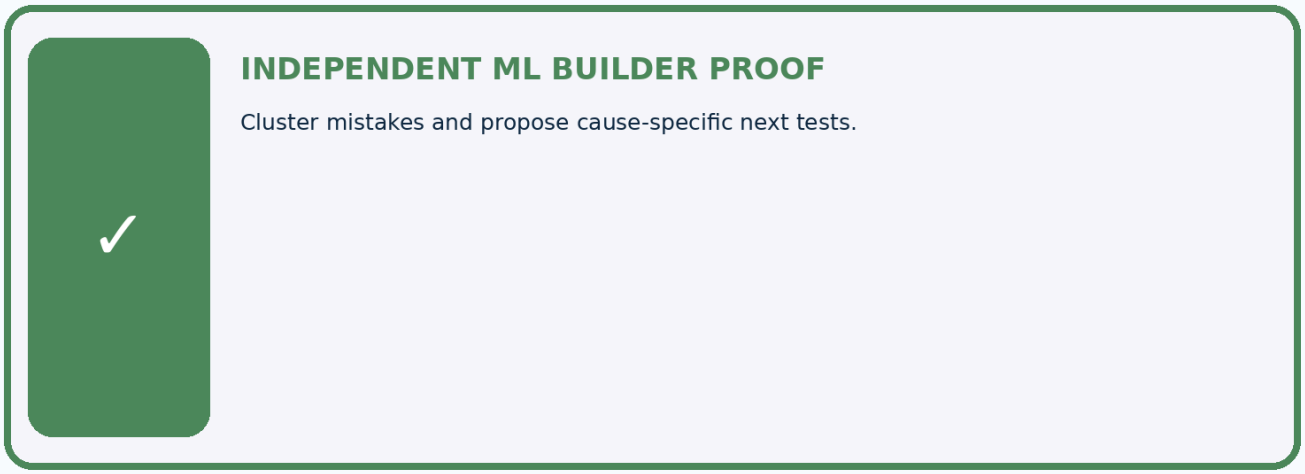
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Cluster mistakes and propose cause-specific next tests.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: **Regression** and Residuals.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Error Analysis case for supported practice. Repeated examples do not become fresh assessments.

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Four errors occur among five backlit photos; two among twenty ordinary photos.

The question

Compare rates and investigate.

Worked explanation

Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare error rates, not only error counts

Evidence or stage	Value or result
Backlit errors	4/5
Ordinary-light errors	2/20

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Error Analysis, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Find an error cluster before choosing a controlled follow-up.

By the end, I can	How I will show it
Categorize development errors using evidence	Explain the deciding step, show a trace or test, and state one limit.
Design a controlled change with a stated hypothesis	Explain the deciding step, show a trace or test, and state one limit.
Preserve failures and test boundaries	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Error Analysis without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Categorize development errors using **evidence**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Find an error cluster before choosing a controlled follow-up.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

1. Plan controlled pair

2. Compute slice errors

3. Inspect dim failures

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan controlled pair Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Read the picture in order.

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Dim errors fall to1/6 in a later same-case replay. Which conclusion is supported by the supplied facts?

1. The replay’s dim error rate is 16.7%; it is development **evidence** if those cases guided the

revision.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

Observed improvement on reused cases is not an untouched final generalisation test.

Repair a missing building block

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

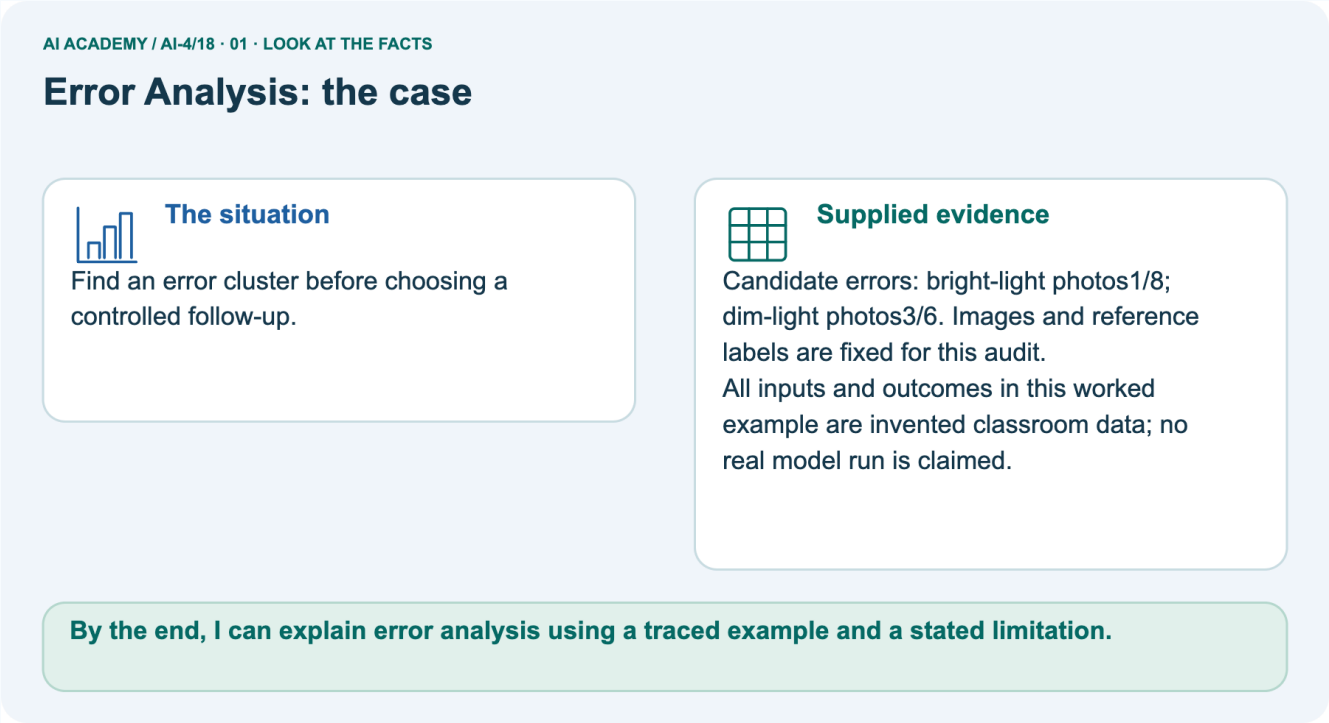
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Error Analysis graphic using its labels, deciding evidence and relevant limitation.

01. Error Analysis: the case



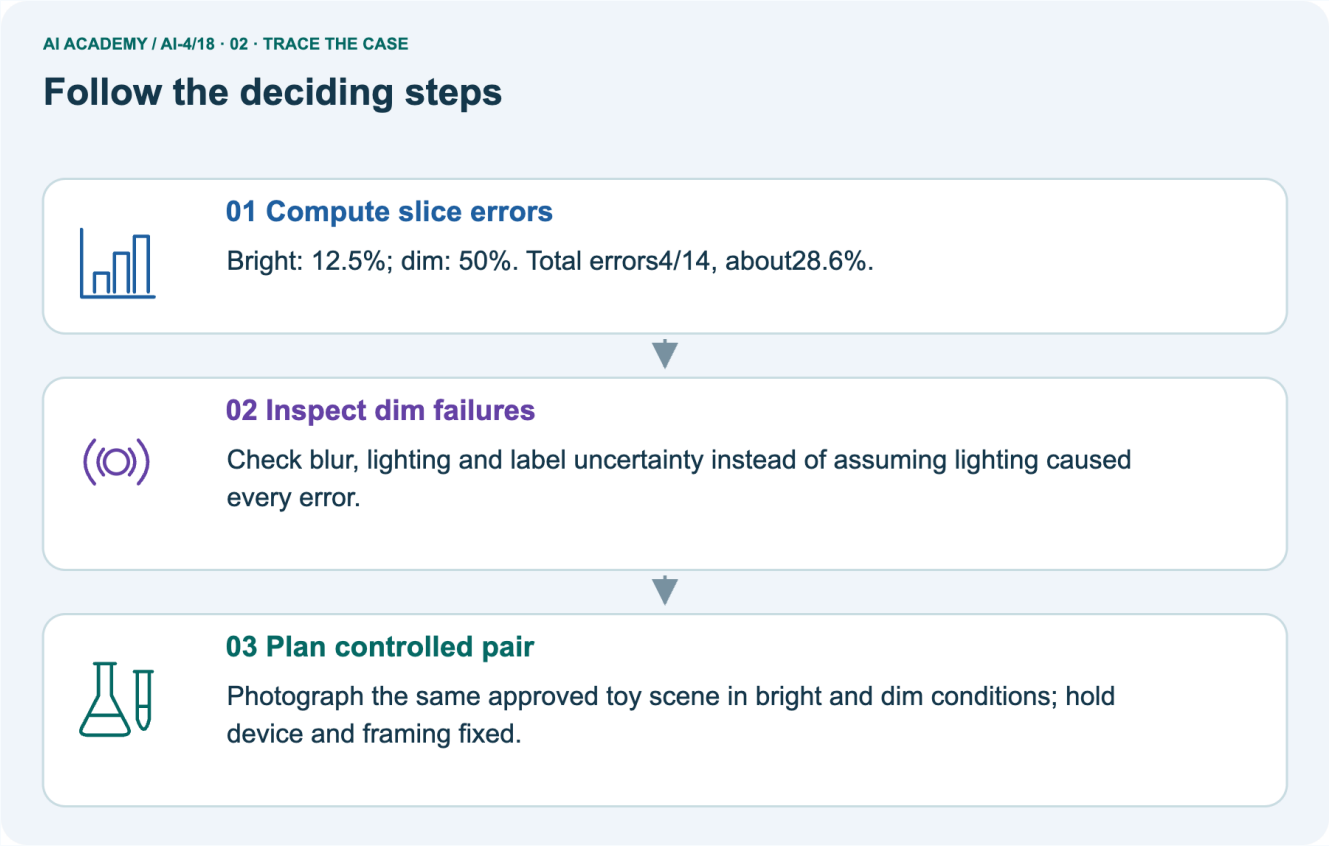
Follow the labelled picture with the matching explanation. Open the vector diagram

Find an error cluster before choosing a controlled follow-up. Supplied facts: Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



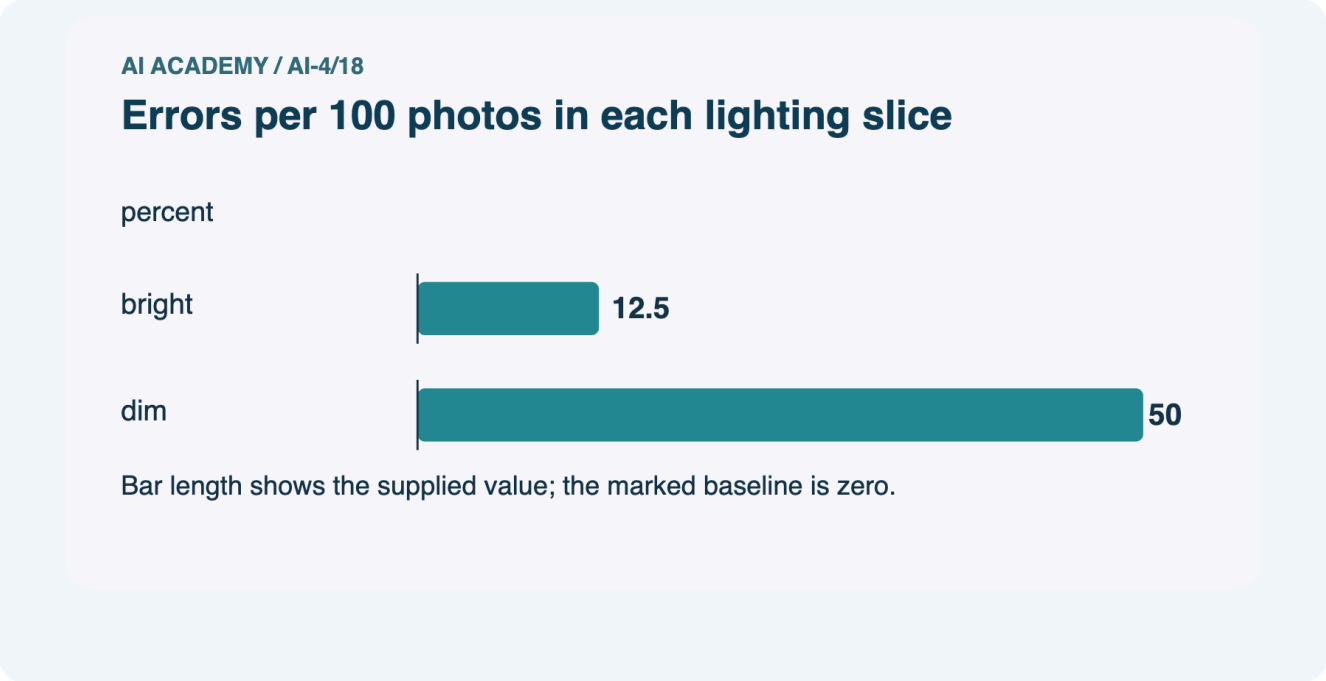
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice errors: Bright: 12.5%; dim: 50%.

Total errors4/14, about28.6%. → Inspect dim failures: Check blur, lighting and label uncertainty instead of assuming lighting caused every error. → Plan controlled pair: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Errors per 100 photos in each lighting slice

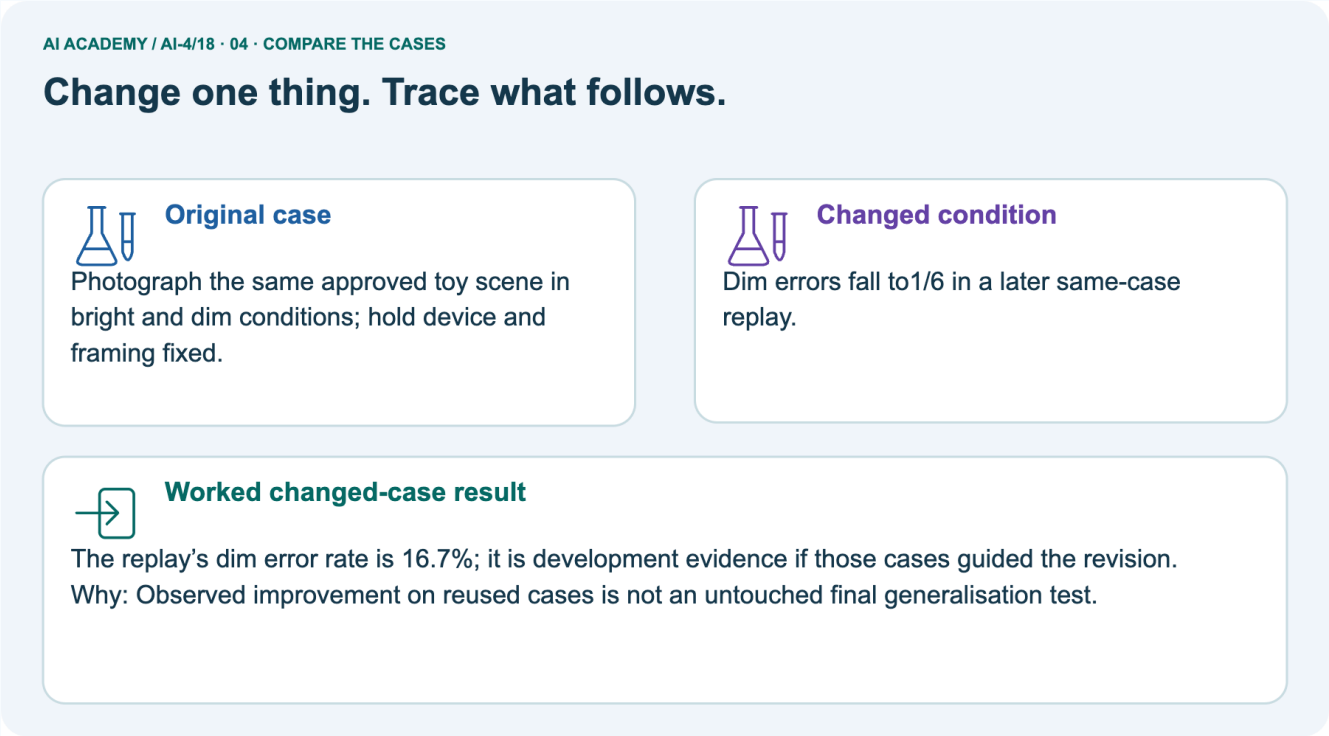


Follow the labelled picture with the matching explanation. Open the vector diagram

Errors per 100 photos in each lighting slice. Units: percent. bright: 12.5. dim: 50.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



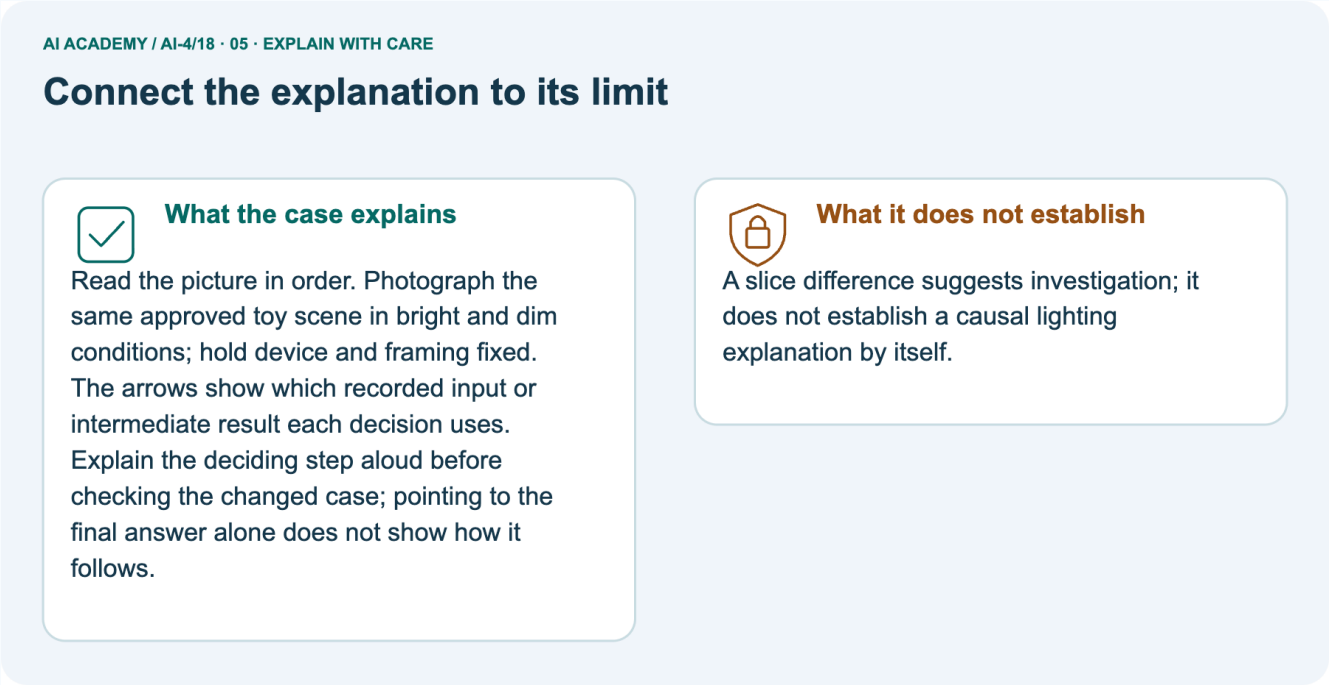
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Changed condition: Dim errors fall to1/6 in a later same-case replay.

Expected result: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Reason: Observed improvement on reused cases is not an untouched final generalisation test.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

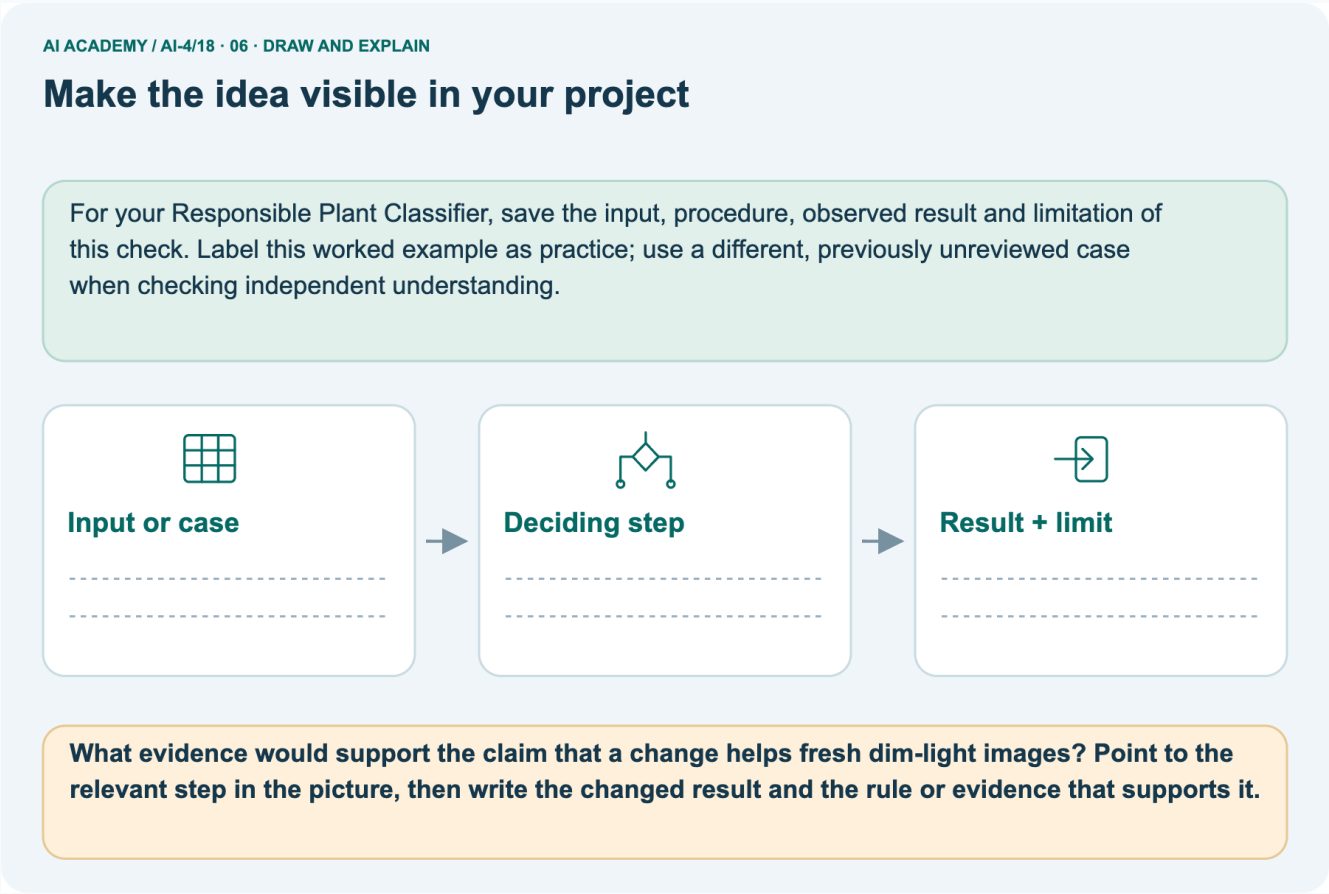
Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.